SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1105

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05 **Total Marks:40**

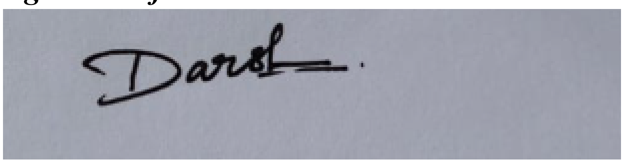
Annexure A

MOCK INTERVIEW

Code of Conduct:

I DARSHINIE KARANI.V.M (192520033) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Mock Interview – Sample Questions (5 Questions)

Mock Interview Test Format for Object Oriented Analysis and Design (OOAD)

Section A	**Self Introduction & Fundamentals**	**3 Minutes**	**5**
Section B	**Technical Questions (OOAD Concepts)**	**7 Minutes**	**10**
Section C	**UML Modeling & Design**	**7 Minutes**	**10**
Section D	**Case Study / Problem Solving**	**8 Minutes**	**10**
Section E	**HR & Career-Oriented Questions**	**5 Minutes**	**5**

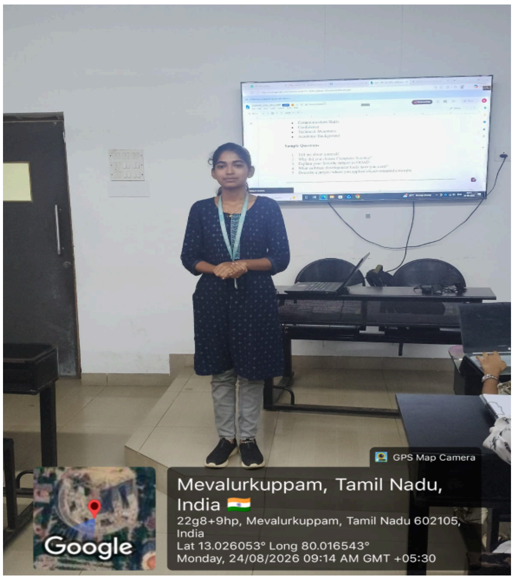
SECTION A – SELF INTRODUCTION (5 Marks)

Evaluation Criteria

- Communication Skills
- Confidence
- Technical Awareness
- Academic Background

Sample Questions

1. Tell me about yourself.
2. Why did you choose Computer Science?
3. Explain your favorite subject in OOAD.
4. What software development tools have you used?
5. Describe a project where you applied object-oriented concepts.



SECTION B – TECHNICAL QUESTIONS (10 Marks)

Basic Level

Q1. What is Object-Oriented Analysis and Design?

Object oriented analysis and design is a concept where we use diagrams for building softwares.

Q2. Differentiate between Analysis and Design.

Analysis: Identifies what the system should do and its requirements.

Design: Decides how the system will be implemented.

Q3. Explain Encapsulation with an example.

Encapsulation means combining data and methods inside a class and protecting the data from direct access. It is usually achieved using other variables and public methods.

Q4. What is Inheritance? Mention its types.

Inheritance is an OOP concept where a child class gets the properties and methods of a parent class.

Q5. Define Polymorphism and its benefits.

Polymorphism means one name having many forms. It allows the same method to behave differently for different objects.

Q6. What is Abstraction?

Abstraction means hiding implementation details and showing only essential features.It can be achieved using abstract classes and interfaces.

Q7. Difference between Aggregation and Composition.

Aggregation: Weak has a relationship; the child can exist independently.

Composition: Strong has a relationship; the child depends on the parent.

Q8. What is Coupling and Cohesion?

Coupling: Dependency between different classes/modules.

Cohesion: They are closely related the responsibilities within one class are.
A good design aims for low coupling and high cohesion.

Q9. What is a Class and an Object?

A class is a blueprint or template for creating objects.An object is an instance of a class.

Q10. Explain Object Responsibility.

Object responsibility means the specific task or duty assigned to an object/class.Proper responsibility assignment makes software clear and maintainable.

SECTION C – UML MODELING (10 Marks)

Q11. What is UML?

UML stands for Unified Modeling Language. It is a standard visual language used to represent, design, and document the structure and behavior of software systems.

Q12. When would you use a Use Case Diagram?

A Use Case Diagram is used to show the functional requirements of a system. It represents actors and the interactions they have with the system.

Q13. Draw and explain a Class Diagram for a Library Management System.

A Class Diagram shows classes, attributes, methods, and relationships. For a library system, classes can include Book, Member, Librarian, and Issue.

Q14. Differentiate Sequence Diagram and Activity Diagram.

A Sequence Diagram shows interactions and the order of messages between objects. An Activity Diagram shows the workflow or sequence of activities in a process.

Q15. Explain State Diagram with an example.

A State Diagram shows the different states of an object and transitions between them. For example, an order can change from New to Confirmed to Shipped to Delivered.

Q16. What is a Deployment Diagram?

A Deployment Diagram shows the physical arrangement of hardware and software in a system. It represents nodes such as servers, computers, and their connections.

Q17. How do Component Diagrams help software architects?

Component Diagrams show the major software components and their dependencies. They help architects understand and plan the high-level structure of a software system.

Q18. What is Multiplicity in UML?

Multiplicity specifies how many objects can participate in a relationship

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SECTION D – CASE STUDY / PROBLEM SOLVING (10 Marks)

Case Study 1

A hospital wants a system to manage:

- Patients
- Doctors
- Appointments
- Billing

Questions

1. Identify the classes.
2. Draw a simple class diagram.
3. Suggest relationships among classes.
4. Which UML diagrams would you use for analysis?
5. Explain how inheritance can be applied.

Answers

1. The main classes are patient, doctors, appointment, and billing. These classes represent the major entities involved in the hospital management system.
2. Patient —— Appointment —— Doctor

   |

   billing

3. A patient books an appointment with a doctor. the Patient is also associated with billing , while the doctor provides the medical consultation.
4. I would only choose use case diagrams to identify actors and requirements, and a class and a class diagram to identify the classes and the relationships.
5. A general person class can be created with common attributes such as name and ID. Patient and doctors can inherit these common properties from the person class

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Case Study 2

An Online Shopping System supports:

- Customers
- Products

- Cart
- Orders
- Payments

Questions

1. Identify actors and use cases.
2. Design a use case diagram.
3. Suggest suitable design patterns.
4. Explain how polymorphism can be used in payment processing.

Answers

1. The main actors are customer, admin and payment system . uses cases includes login searching for the product, adding to the cart, place order, making payment , manage products.
2. CUSTOMER – Login

   – Search the product

   – Add to cart

   – place the order

   – make payment

   Admin– Manage products

3. The factory pattern can be used to create different payment objects, while strategy pattern supports different payment methods. These patterns make the system flexible and easier to extend.
4. A common payment interface can have implementations such as Credit card payment , upi payment , netbanking payment. The same pay method can behave differently for each payment type.

SECTION E – HR & PROFESSIONAL QUESTIONS (5 Marks)

Q19. Why should we hire you?

You should hire me because i am quick learner and responsible and willing to improve my technical skills . and i’m ready to face new challenges.

Q20. What are your strengths and weaknesses?

My strengths arequick learning , team work , and problem solving. My weaknesses is that i sometimes spend extra time making sure my work is correct or not, but i’m learning too manage time better.

Q21. How do you handle project deadlines?

i handle my deadlines by planning my tasks , setting priorities and completing important work first. I can communicate with my team if i face any difficulty or delay.

Q22. Have you worked in a team project? Explain your role.

Yes, i have worked in team projects where we contribute by sharing ideas , completing assigned tasks, and coordinating with team members. i have also helped my team when they faced any difficulties in them

Q23. Where do you see yourself in five years?

In five years, i see myself as a skilled and confident professional in the technology field . and i want to gain strong technical knowledge , work on real world projects and continuously improve my skills.

RUBRICS FOR CALCULATION:

MOCK INTERVIEW

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately